

TryHackMe – Linux Fundamentals

1. Introduction

Linux is an open-source operating system that is widely used in **servers, websites, cybersecurity, mobile devices, industrial systems, and other computing devices**. In cybersecurity, Linux is very important because many security tools and tasks are performed through the **terminal**.

2. Linux Terminal and Basic Commands

In this lab, I used an Ubuntu Linux machine provided by TryHackMe. The terminal allows users to interact with the system by entering commands.

Some basic commands practiced in the lab were:

Command Purpose

whoami Shows the current user

echo Displays text

ls Lists files and folders

cd Moves to another directory

cat Displays file contents

pwd Shows the current working directory

find Searches for files

grep Searches for specific text

For example, whoami was used to identify the current user, while pwd was used to check the current location in the file system.

3. File and Text Searching

Linux provides commands to quickly search through files. The find command can search for a file by its name, while grep searches for specific text inside a file. In the lab, the grep command was used to search the access.log file: `grep "THM" access.log`

This helped locate the flag present inside the log file.

4. Linux Operators

The lab also introduced operators used to combine commands and redirect output.

Operator	Purpose
&	Runs a command in the background
&&	Runs the next command after the first finishes
>	Sends output to a file and overwrites existing content
>>	Adds output to the end of a file

For example:

```
echo "TryHackMe" > thm
```

This creates a file named thm and stores the text **TryHackMe** in it.

Using:

```
echo "TryHackMe" >> thm
```

adds the text to the existing file instead of replacing its contents.

5. Practical Work Completed

During this TryHackMe lab, I practiced:

- Identifying the current Linux user.
- Displaying text using echo.
- Navigating directories using ls, cd, and pwd.
- Reading files using cat.
- Searching for files using find.
- Searching file contents using grep.
- Using operators such as &, &&, >, and >>.
- Redirecting command output into files.

6. Conclusion

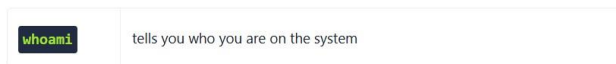
This lab provided a basic understanding of **Linux and its terminal commands**. I learned how to navigate the Linux file system, search files and text, and redirect command output. These basic Linux skills are useful for **cybersecurity, system administration, and security investigations**.

Questions given in the room:

1) What command would we use to find out who we are on the system?

Whoami

2) What command would we use to output some text? Echo



When you run your first command, you will see some text. We call this output. Interacting with Linux feels like a conversation. You give it an instruction, and it gives you output. We can ask Linux to output specific text for us.



To echo the text "TryHackMe", we can use `echo TryHackMe`. For multiple words, we will need to wrap these in quotes. For example, `echo "hello world"`. Using `echo` is just the start of a very important skill in Cyber security: being able to create output and sending it somewhere. You will get to see how this skill is further used in the upcoming tasks.

3) Run `ls` in the current folder. How many folders are there?

4

4)

One of those folders contains a file. Which folder is it? folder1

5)

Use `cat` to read the file within this folder. What does it say? password123

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Thu Jul 23 15:11:29 2026 from 10.81.103.223
tryhackme@linux1:~$ ls
access.log  folder1  folder2  folder3  folder4
tryhackme@linux1:~$ ls folder1
access.log  passwords.txt
tryhackme@linux1:~$ cd folder1
tryhackme@linux1:~/folder1$ cat password.txt
cat: password.txt: No such file or directory
tryhackme@linux1:~/folder1$ cat passwords.txt
password123
tryhackme@linux1:~/folder1$
```

6)After using grep THM access.log a flag was found. What is the flag? THM{ACCESS}

```
tryhackme@linux1:~/folder1$ cd /home/tryhackme
tryhackme@linux1:~$ grep "THM" access.log
13.127.130.212 - - [04/May/2021:08:35:26 +0000] "GET THM{ACCESS} lang=en HTTP/1.1" 404 360 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/77.0.3865.120 Safari/537.36"
tryhackme@linux1:~$
```

7)What operator waits for the first command to finish before running the next command?

&&

8)What redirector allows us to save the output of a command **without overwriting** the contents of a file?

>>

	Runs the command, but does not wait for it to finish before you can do anything else. The command runs in the background, and is helpful for commands that might take a while to complete, or ones that you want to keep running.
	Runs both commands, but waits for the first command to finish first, before the next. Like a set of dominoes.
	Used to redirect output. We can take the output of a command and send it to a file. This operator will overwrite anything that exists in the file.
	This redirector does the same thing, but instead of overwriting, it will just add the output to the bottom of the file.